

# Zan Ahmad

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## Education

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| <b>Johns Hopkins University</b> , Whiting School of Engineering<br>Ph.D. Candidate, Applied Mathematics & Statistics<br>Advisors: <a href="#">Natalia Trayanova</a> , <a href="#">Mauro Maggioni</a>                       | June 2022 – May 2027<br>(expected) |
| <b>Johns Hopkins University</b> , Whiting School of Engineering<br>M.S.E., Applied Mathematics & Statistics                                                                                                                | June 2022 – May 2025               |
| <b>New York University</b> , Courant Institute of Mathematical Sciences<br>B.A., Mathematics (Honors)<br>Minors: Chemistry, Film Production<br>Advisors: <a href="#">Charles S. Peskin</a> , <a href="#">Charles Puelz</a> | Sept 2017 – Dec 2020               |

## Research Interests

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Geometric machine learning · Riemannian shape analysis · Neural operators for PDEs · Cardiovascular modeling

## Research Experience

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| <b>Johns Hopkins University</b> , Institute of Computational Medicine<br>Graduate Researcher, <a href="#">Computational Cardiology Lab</a><br>Advisors: <a href="#">Natalia Trayanova</a> , <a href="#">Mauro Maggioni</a>                       | June 2022 – Present  |
| <b>Yale School of Medicine</b> , Department of Neurology<br>Postgraduate Researcher, <a href="#">Blumenfeld Lab</a><br>Advisor: <a href="#">Hal Blumenfeld</a>                                                                                   | June 2021 – May 2022 |
| <b>New York University</b> , Center for Neural Science<br><a href="#">NIH Training Program in Computational Neuroscience</a><br>Advisor: <a href="#">Charles S. Peskin</a>                                                                       | Sept 2020 – May 2021 |
| <b>New York University</b> , Courant Institute of Mathematical Sciences<br><a href="#">Applied Mathematics Summer Undergraduate Research Experience (AM-SURE)</a><br>Advisors: <a href="#">Charles Puelz</a> , <a href="#">Charles S. Peskin</a> | June 2020 – Aug 2020 |
| <b>New York University</b> , Courant Institute of Mathematical Sciences<br><a href="#">Modeling &amp; Simulation Research Training Group</a><br>Advisor: <a href="#">Charles Puelz</a>                                                           | Sept 2019 – May 2020 |
| <b>New York University</b> , Department of Chemistry<br>Undergraduate Researcher, <a href="#">DNA Nanotechnology Lab</a><br>Advisors: <a href="#">Nadrian Seeman</a> , <a href="#">Yoel P. Ohayon</a>                                            | Jan 2018 – Aug 2019  |

## Publications & Preprints

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<sup>†</sup>equal contribution. [Full list on Google Scholar.](#)

### Published

1. “A Computational Pipeline for Clustering Left Atrial Appendage Morphology via Elastic Shape Analysis.”  
**Z. Ahmad**, M. Yin, Y. Sukurdeep, N. Rotenberg, E. Kholmovski, N. Trayanova.  
*Computers in Biology and Medicine*, 196:110905, 2025.
2. “A Comprehensive Stroke Risk Assessment by Combining Atrial Computational Fluid Dynamics Simulations and Functional Patient Data.”  
A. Zingaro, **Z. Ahmad**, E. Kholmovski, K. Sakata, L. Dede’, A.K. Morris, A. Quarteroni, N. Trayanova.  
*Scientific Reports*, 14:9515, 2024.
3. “Optimal Fenestration of the Fontan Circulation.”  
**Z. Ahmad**, L.H. Jin, D.J. Penny, C. Rusin, C.S. Peskin, C. Puelz.  
*Frontiers in Physiology: Computational Physiology and Medicine*, 13, 2022.

### Workshop Papers

1. “Diffeomorphic Latent Neural Operators for Data-Efficient Learning of Solutions to PDEs.”  
**Z. Ahmad**, S. Chen, M. Yin, A. Kumar, N. Charon, N. Trayanova, M. Maggioni.  
*AAAI 2025 Workshop on AI for Accelerating Science and Engineering*, 2024.
2. “Convolutional Deep Operator Networks for Predicting Nonlinear Wave Propagation in Heterogeneous Spinal Cord Anatomy.”  
A. Kumar, X. Zhi, **Z. Ahmad**, M. Yin, A. Manbachi.  
*AAAI 2025 Workshop on AI for Accelerating Science and Engineering*, 2024.

### Under Review & Preprints

1. “Predicting the *When*: Multimodal AI for Time-to-Recurrence Analysis After Atrial Fibrillation Ablation.”  
M. Yin, C. Lai, R. Yadav, . . . , **Z. Ahmad**, S. Chen, . . . , N. Trayanova.  
*medRxiv preprint*, 2026.
2. “Stimulation of the Thalamus for Arousal Restoral in Temporal Lobe Epilepsy (START) Clinical Trial: Modulation of Natural Sleep and Focal Seizures.”  
T. Yadav, Z. Zhang, V. Kremen, **Z. Ahmad**, . . . , H. Blumenfeld.  
*medRxiv preprint*, 2025.
3. “Low and High Frequency Signatures of Impaired Consciousness in Temporal Lobe Seizures.”  
T. Yadav, . . . , **Z. Ahmad**, . . . , H. Blumenfeld.  
*bioRxiv preprint*, 2025.
4. “Steady-State Analysis of Gravitational Effects on Hemodynamics.”  
A. Kennard<sup>†</sup>, **Z. Ahmad**<sup>†</sup>, R.S. Blue, C.S. Peskin, K.M. Ong.  
*Submitted to Scientific Reports*, 2025.
5. “Thinner Latent Spaces: Detecting Dimension and Imposing Invariance Through Conformal Autoencoders.”  
G. Kevrekidis, **Z. Ahmad**, S. Villar, M. Maggioni, Y. Kevrekidis.  
*Submitted to SIAM Journal on Mathematics of Data Science*, 2024.
6. “Graph Fourier Neural Kernels (G-FuNK): Learning Solutions of Nonlinear Diffusive Parametric PDEs on Multiple Domains.”  
S. Loeffler, **Z. Ahmad**, S.Y. Ali, C. Yamamoto, D.M. Popescu, A. Yee, Y. Lal, N. Trayanova, M. Maggioni.  
*arXiv preprint*, 2024.

### In Preparation

1. “Toward a mechanistic understanding of control during simulated blood loss using compartmental models and time-frequency analysis.”  
C. Puelz, B.E. Griffith, J.L. Marzuola, **Z. Ahmad**, C.S. Peskin, H.-T. Wu.  
*Manuscript in preparation.* [[Password-protected manuscript for hiring committees](#)]

## Teaching

## Teaching portfolio and materials

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| Probability & Statistics for Life Sciences<br><i>Four-credit departmental course; lectures, office hours, and exam grading within a shared curriculum</i><br><a href="#">Materials ↗</a>                                                                                                                                                        | F 2026    | JHU  | Instructor of Record             |
| Introduction to Scientific Programming in Python<br><i>Scientific libraries, numerical linear algebra, optimization, ML (neural networks, CNNs, transformers, LLMs); algorithmic-thinking and NumPy assignment</i><br><a href="#">Assignment ↗</a> · <a href="#">Materials ↗</a>                                                                | W 2026    | JHU  | Instructor of Record             |
| Biomedical Engineering Innovation & Design                                                                                                                                                                                                                                                                                                      | Su 2025   | JHU  | Instructor of Record             |
| Biomedical Engineering Innovation & Design                                                                                                                                                                                                                                                                                                      | Sp 2025   | JHU  | Instructor of Record             |
| Introduction to Scientific Programming in Python<br><a href="#">Materials ↗</a>                                                                                                                                                                                                                                                                 | W 2025    | JHU  | Instructor of Record             |
| Advanced Topics in Computational Cardiology<br><i>0D/1D modeling, shape analysis, manifold learning, computer vision, neural operators; upper-level seminar. Heart-rate feedback-control project; finite-difference Python tutorial</i><br><a href="#">Feedback control ↗</a> · <a href="#">Python tutorial ↗</a> · <a href="#">Materials ↗</a> | F 2024    | JHU  | Creator;<br>Instructor of Record |
| Mathematical Modeling & Computer Simulation<br><i>First-principles biological modeling; course projects served as launching points for undergraduate research mentorship</i><br><a href="#">Materials ↗</a>                                                                                                                                     | F 2024    | JHU  | Creator;<br>Instructor of Record |
| Biomedical Engineering Innovation & Design                                                                                                                                                                                                                                                                                                      | F 2024    | JHU  | Instructor of Record             |
| Biomedical Engineering Innovation & Design                                                                                                                                                                                                                                                                                                      | Su 2024   | JHU  | Instructor of Record             |
| Mathematical Cardiology<br><i>First-principles cardiovascular modeling as an accessible on-ramp to research in mathematical biology and cardiovascular modeling</i><br><a href="#">Materials ↗</a>                                                                                                                                              | F 2023    | JHU  | Creator;<br>Instructor of Record |
| Biomedical Engineering Innovation & Design                                                                                                                                                                                                                                                                                                      | Su 2023   | JHU  | Instructor of Record             |
| Computational Neuroscience: Channels & Networks<br><i>Developed MATLAB tutorials on neuronal action potential models; students used these as a foundation for final course projects. [tutorial videos]</i><br><a href="#">Materials ↗</a>                                                                                                       | Sp 2021   | NYU  | Guest Lecturer                   |
| Linear Algebra                                                                                                                                                                                                                                                                                                                                  | F 2020    | NYU  | Grader                           |
| Calculus I                                                                                                                                                                                                                                                                                                                                      | Sp 2020   | NYU  | Grader                           |
| Calculus I                                                                                                                                                                                                                                                                                                                                      | F 2019    | NYU  | Grader                           |
| Thermodynamics; Organic Chemistry I & II<br><i>F 2019: Thermodynamics and Organic Chemistry II;<br/>Sp 2019: Organic Chemistry I</i>                                                                                                                                                                                                            | Sp–F 2019 | NYU  | Lead Chemistry Tutor             |
| Physics<br><i>East Harlem Tutorial Program: nonprofit providing free tutoring and mentorship to K–12 students in East Harlem, NY</i>                                                                                                                                                                                                            | 2018–19   | EHTP | Volunteer Tutor                  |

## Mentoring

**Shiyi Chen** · M.S. student, Mechanical Engineering, Johns Hopkins University      Apr 2024 – Aug 2025  
 Thesis: [Learning PDE solution operators via diffeomorphic mappings: applications in fluid dynamics](#).  
 Next position: Ph.D. student, Biomedical Engineering, Yale University.

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| <b>Ayush Mandal</b> · Undergraduate, Applied Mathematics & Computational Neuroscience, JHU<br>Topics: Feedback control of heart rate in zero-dimensional circulation models; modeling neuronal dynamics.                                         | Sept 2023 – Nov 2025 |
| <b>Noam Rottenberg</b> · Undergraduate, Engineering, Johns Hopkins University<br>Topics: Automatic left atrial segmentation and preprocessing for statistical shape analysis.<br><i>Next position: Medical student, Yale School of Medicine.</i> | Aug 2023 – Jan 2024  |

## Industry Experience

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| <b>Intuit Inc.</b> , AI/Data Science Department · San Diego, CA<br>AI Science Intern, Consumer Group Analytics and Decision Science<br>Advisor: Dr. Joseph Powers | June 2025 – Sept 2025 |
| <b>Virgin Galactic</b> · Remote<br>Computational Research Intern<br>Advisor: Dr. Karen Ong                                                                        | Jan 2023 – Aug 2023   |

## Grants, Awards & Honors

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| <b>American Heart Association Predoctoral Fellowship</b> · \$67,388<br><i>Quantifying Stroke Risk in Atrial Fibrillation with Computational Fluid Dynamics and Machine Learning.</i>                                              | July 2024 – June 2026 |
| <b>IEEE EMBS Student Fellowship</b> · \$2,000<br>One of 14 selected from 900 applicants. Full scholarship for AI in Medicine & Healthcare workshop. Chania, Crete, Greece.                                                        | June 2024             |
| <b>IDIES Poster Competition — 1st Place</b> · \$100<br><i>Random Convolutional Features and Patch-Based Learning for Multitask Image Classification.</i>                                                                          | Oct 2023              |
| <b>NIH Cardiovascular Epidemiology Training Grant</b> (T32HL007024) · \$27,400<br>Fellowship for PhD students with research focused on cardiovascular disease.                                                                    | July 2023 – June 2024 |
| <b>Acheson J. Duncan Fund Travel Award</b> · \$3,150<br>Fund for the Advancement of Research in Statistics.                                                                                                                       | April 2023            |
| <b>NIH Computational Medicine Training Grant</b> (T32GM119998) · \$25,836<br>Fellowship for PhD students at the Institute of Computational Medicine, JHU.                                                                         | June 2022 – May 2023  |
| <b>NIH Computational Neuroscience Training Grant</b> (R90DA043849) · \$13,336<br>Awarded to 6 of 50 applicants. <i>Short time-scale dynamics in a computational model of signal propagation in a cerebellar neuronal network.</i> | Aug 2020 – May 2021   |
| <b>NSF Modeling &amp; Simulation RTG</b> (DMS-1646339) · \$6,250<br>AM-SURE at NYU Courant Institute. Three undergraduates selected from 60 applicants.                                                                           | May – Aug 2020        |
| <b>Wilfred L. &amp; Ruth S.F. Peltz Research Scholar Award</b> · \$2,750<br>Awarded for NYU DURF proposal: <i>Optimally fenestrated Fontan circulation.</i>                                                                       | May 2020              |
| <b>Dean's Undergraduate Research Fund Grant (DURF)</b> · \$1,000<br><i>Optimal fenestration of Fontan circulation.</i> NYU College of Arts and Sciences.                                                                          | May 2020              |
| <b>Dean's Undergraduate Research Fund Training Grant</b> · \$500<br><i>Self-assembly of A-form 3D DNA crystals of 2'-fluoro DNA.</i>                                                                                              | May 2018              |

## Seminars & Invited Talks

1. “A Scalable Approach to Riemannian Geometric Statistics for Cardiovascular Shape Modeling.”  
*Mostly Biomathematics Lunchtime Seminar*, April 2026.
2. “Neural Operators for Learning PDEs on Geometrically Varying Data.”  
*JHU Applied Mathematics & Statistics Department Seminar*, April 2024 & December 2024.
3. “Hemodynamic Indicators of Stroke Risk Uncovered via Personalized Fluid Dynamics Simulations.”  
*JHU Applied Mathematics & Statistics Department Seminar*, September 2023.

4. “Steady-State Analysis of Gravitational Effects on Hemodynamics.”  
*JHU Applied Mathematics & Statistics Department Seminar*, February 2023.
5. “Pulsatile Model for the Fontan Circulation and a Feedback Control Method for Heart Rate.”  
*JHU Applied Mathematics & Statistics Department Seminar*, November 2022.
6. “Short Time-Scale Dynamics in a Computational Model of Signal Propagation in a Cerebellar Neural Network.”  
*NYU Training Program for Computational Neuroscience Conference*, June 2021.
7. “Using Mathematics to Study and Optimize Congenital Heart Disorders.”  
*NYU College of Arts and Sciences Undergraduate Research Conference*, May 2021.
8. “Feedback Control of Blood Pressure and Heart Rate in a Pulsatile Cardiovascular Model.”  
*NYU Courant Institute Modeling and Simulation Research Training Group Seminar*, March 2021.
9. “Mathematical Model for an Optimally Fenestrated Fontan Circulation.”  
*NYU Courant Institute Applied Mathematics Research Presentations*, October 2020.

## Conference Presentations

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1. “AI-Based Shape Analysis Correlates Left Atrial Appendage Morphology with Stroke Risk in Patients with Atrial Fibrillation.”  
*Heart Rhythm Society (HRS)*, San Diego, CA, April 2025.
2. “Multimodal Artificial Intelligence for Predicting Atrial Fibrillation Recurrence Using Pre-Procedural Data.”  
*Heart Rhythm Society (HRS)*, San Diego, CA, April 2025.
3. “Mathematical Cardiology: Operator Learning, Shape Analysis, and Computational Fluid Dynamics.”  
*IEEE EMBS Leadership Academy Workshop on AI/Data Science in Medicine and Healthcare*, Chania, Crete, Greece, June 2024.
4. “Random Convolutional Features and Patch-Based Learning for Multitask Image Classification.”  
*IDIES Annual Research Symposium*, Baltimore, MD, October 2023. **1st Place Best Poster Award.** [\[poster\]](#)
5. “Random Convolutional Features and Patch-Based Learning for Multitask Image Classification.”  
*Johns Hopkins University AI-X Symposium*, Baltimore, MD, September 2023.
6. “Non-Pulsatile ODE Model of Circulation Response to Hypergravity.”  
*SIAM Conference on Control Theory*, Philadelphia, PA, July 2023.
7. “Modeling Feedback Control Response to Hemodynamic Perturbations in a Pulsatile Circulation Model.”  
*SIAM Conference on Control Theory*, Philadelphia, PA, July 2023.
8. “Hemodynamics and Oxygen Model for the Fontan Circulation with Feedback Control in Variable Heart Rate.”  
*ECCOMAS Minisymposium on Multiphysics and Multiscale Simulations in Cardiovascular Applications*, Porto, Portugal, June 2023.
9. “A Dynamic Mathematical Model for the Prediction of Cardiovascular Response to Hypergravity.”  
*American Society of Aerospace Medicine (ASMA)*, New Orleans, LA, May 2023.
10. “Stroke Risk Is Identified by Slow Blood Flow in the Left Atrium.”  
*Heart Rhythm Society (HRS)*, New Orleans, LA, May 2023.
11. “Modeling the Interplay Between Blood Flow and Oxygen Transport in the Defective Heart.”  
*SIAM Conference on Dynamical Systems*, New Orleans, LA, May 2023.
12. “Hemodynamic Indicators of Stroke Risk Uncovered by Personalized Fluid Dynamics Models.”  
*Cardiac Physiome Workshop*, Irvine, CA, May 2023.
13. “A Dynamic Model for the Control of Cardiovascular Response to Spaceflight.”  
*International Conference of Aerospace Medicine*, Paris, France, September 2022.
14. “Simulation of Circulation Response to Gravitational Changes During Spaceflight.”  
*SIAM Conference on Life Sciences*, Pittsburgh, PA, July 2022.
15. “Optimal Fenestration of the Fontan Circulation.”  
*SIAM Annual Meeting*, Pittsburgh, PA, July 2022. [\[report\]](#)
16. “Mathematical Modeling of Cardiac Circulation Under Hypergravity.”  
*American Society of Aerospace Medicine (ASMA)*, Reno, NV, May 2022.
17. “Increased Intracranial EEG Power and Duration in Temporal Lobe Seizures with Impaired Consciousness.”  
*American Epilepsy Society*, Chicago, IL, December 2021.

18. “Mechanisms of Impaired Consciousness in Medial Temporal Lobe Seizures Investigated with Intracranial EEG.”  
*Society for Neuroscience (SfN)*, Chicago, IL, November 2021.
19. “Mathematical Model for an Optimally Fenestrated Fontan Circulation.”  
*Undergraduate Mathematics Symposium, University of Illinois at Chicago*, Chicago, IL, December 2021.

## Service & Leadership

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- Founder and President, **JHU SIAM Student Chapter**. Lead organizer of inaugural SIAM DMV Area Applied Math Conference, April 2025 — full-day event with UMDCP, UMBC, GWU, GU, GMU, and JHU, over 100 attendees.
- Editor-in-Chief, **NYU Medical Dialogue Review**.
- Volunteer Physics Tutor, **East Harlem Tutorial Program (EHTP)** — January 2019 Volunteer of the Month.
- Volunteer Tutor, **The Perfect Tutor** (NYU non-profit) — Lead Math Tutor.

## Media Publications

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1. [“Circulation Models Assess the Impacts of Congenital Heart Defects and Hypergravity.”](#)  
Jillian Kunze, *SIAM News*, August 2023. *Coverage of SIAM Control Theory Conference presentation.*
2. [“Medicine and Biology Through the Lens of Mathematics.”](#)  
**Z. Ahmad**, *NYU Medical Dialogue Review*, Vol. XIV, Issue 2, pp. 44–47, April 2020.